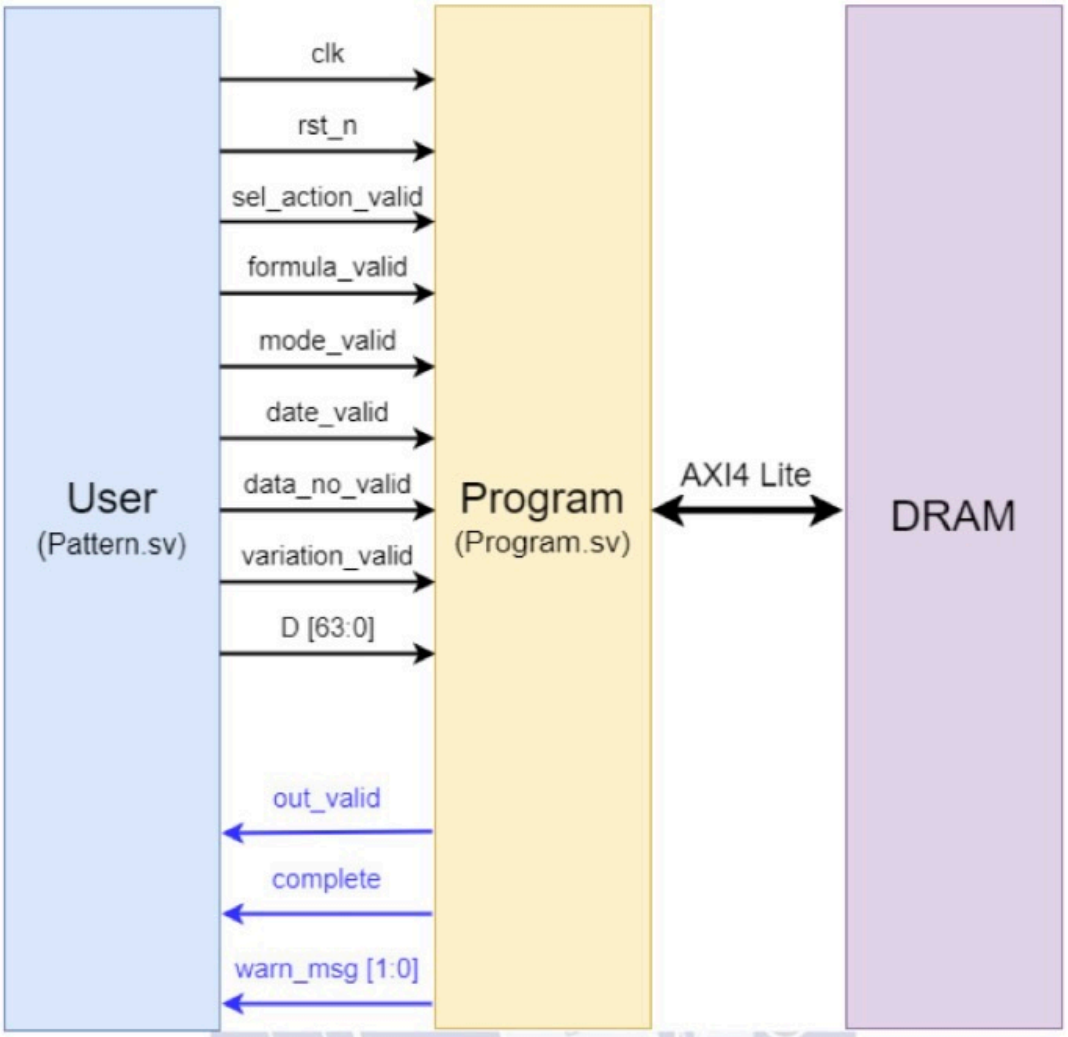


Figure 1: The Design Block Diagram

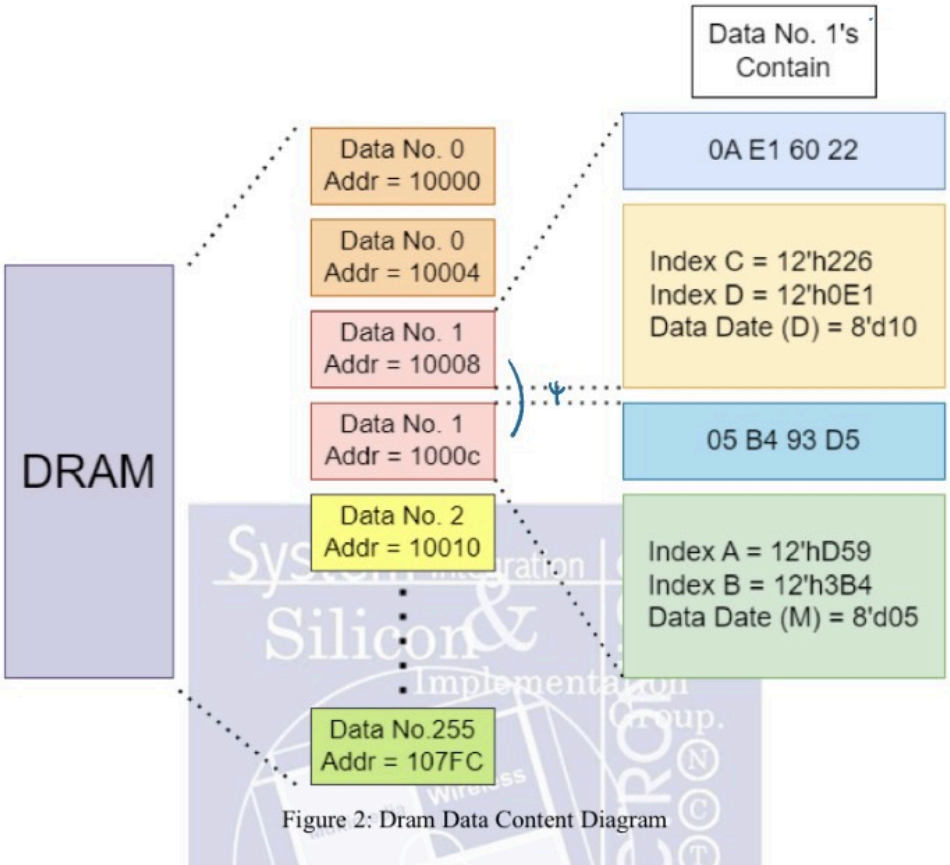


Figure 2: Dram Data Content Diagram

Table 4: Formula

Formula	Input	Content
A	3'd0	$R = \left\lfloor \left(\frac{I(A) + I(B) + I(C) + I(D)}{4} \right) \right\rfloor$
B	3'd1	$R = max(I(A), I(B), I(C), I(D)) - \min(I(A), I(B), I(C), I(D))$
C	3'd2	$R = \min(I(A), I(B), I(C), I(D))$
D	3'd3	$R = 1_{\{I(A) \geq 2047\}} + 1_{\{I(B) \geq 2047)\}} + 1_{\{I(C) \geq 2047\}} + 1_{\{I(D) \geq 2047\}}$ $Ex: I(A) = I(B) = 0, I(C) = I(D) = 4095, R = 2$
E	3'd4	$R = 1_{\{I(A) \geq TI(A)\}} + 1_{\{I(B) \geq TI(B)\}} + 1_{\{I(C) \geq TI(C)\}} + 1_{\{I(D) \geq TI(D)\}}$
F	3'd5	$n = argmax_{n \in \{A,B,C,D\}}(G(n))$, $R = floor(\frac{1}{3}(\sum_{i \neq n, i \in \{A,B,C,D\}} G(i)))$ $Ex: G(A) = G(B) = G(C) = 3, G(D) = 5,$ $R = \frac{G(A) + G(B) + G(C)}{3} = 3$
G	3'd6	$N_0, N_1, N_2, N_3 = ascended.sort(G(A), G(B), G(C), G(D))$, $R = \left\lvert \left(\frac{N_0}{2}\right) \right\rvert + \left\lvert \left(\frac{N_1}{4}\right) \right\rvert + \left\lvert \left(\frac{N_2}{4}\right) \right\rvert$ $Ex: G(A) = 5, G(B) = G(C) = 4, G(D) = 2$ $N_0, N_1, N_2, N_3 = 2,4,4,5 \;\; R = 3$
H	3'd7	$R = \left\lfloor \frac{G(A) + G(B) + G(C) + G(D)}{4} \right\rfloor$

Note: R means result. I(A), I(B), I(C), I(D) are early trading indices A, B, C, D from dram respectively. TI(A), TI(B), TI(C), TI(D) are today's late trading session indices A, B, C, D respectively. G(A), G(B), G(C), G(D) are $|I(A) - TI(A)|$, $|I(B) - TI(B)|$, $|I(C) - TI(C)|$, $|I(D) - TI(D)|$ respectively. If $R \geq Threshold$, you must issue the warning message.

Table 5: The corresponding value of actions and warning messages

ACTION		WARNING MESSAGE	
Index Check	2'b00	Date_Warn	2'b01
		Risk_Warn	2'b10
Update	2'b01	Data_Warn	2'b11
Check Valid Date	2'b10	Date_Warn	2'b01
All action		No_Warn	2'b00

Input: Program.sv

Signal	Width	from	Note
clk	1	testbench	System clock
rst_n	1	pattern	Asynchronous reset active low reset. Every output signal should be zero after **rst_n**.
sel_action_valid	1	pattern	High when input means select the action.
formula_valid	1	pattern	High when input means type of formula.
mode_valid	1	pattern	High when input means the Mode.
date_valid	1	pattern	High when input means today's date or expired date.
data_no_valid	1	pattern	High when input means the number of data.
index_valid	1	pattern	High when input means variation (Will pull HIGH 4 times)
D[71:0]	72	pattern	Represents the contents of the current input. = {70'bX, Action} = {69'bX, Formula_Type} = {70'bX, Mode} = {63'bX, Month[3:0], Day[4:0]} = {64'bX, Data No.} = {60'bX, Index or Variation}

Output: Program.sv

Signal	Width	Send to	Note
out_valid	1	pattern	Should set to high when your output is ready. **out_valid** will be high for **only one** cycle.
warn_msg	2	pattern	warn_msg will be 2'b00 (No warn) if operation is complete, else it needs to be corresponding value.
complete	1	pattern	1'b1: operation complete 1'b0: some warnings occurred
AR_VALID	1	DRAM	AXI Lite signal
AR_ADDR	17	DRAM	AXI Lite signal
R_READY	1	DRAM	AXI Lite signal
AW_VALID	1	DRAM	AXI Lite signal
AW_ADDR	17	DRAM	AXI Lite signal
W_VALID	1	DRAM	AXI Lite signal
W_DATA	64	DRAM	AXI Lite signal
B_READY	1	DRAM	AXI Lite signal
AR_READY	1	DRAM	AXI Lite signal
R_VALID	1	DRAM	AXI Lite signal
R_DATA	64	DRAM	AXI Lite signal
R_RESP	2	DRAM	AXI Lite signal
AW_READY	1	DRAM	AXI Lite signal
W_READY	1	DRAM	AXI Lite signal
B_VALID	1	DRAM	AXI Lite signal
B_RESP	2	DRAM	AXI Lite signal